5DGSPTSBL: Geometric Fast-Slow Stable Boundary Layer Model

5DGSPTSBL Contributors

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Abstract

We present a five-dimensional Geometric Singular Perturbation Theory (GSPT) framework for the atmospheric stable boundary layer (SBL) with state vector $\mathbf{x} = (\tilde{e}, q_\theta, S, T_s, T_g)^T$. By introducing a C^∞-smooth chart desingularization ($\tilde{e} = \sqrt{e+\delta}$), we resolve the non-smooth turbulent kinetic energy singularity at $e = 0$, enabling formal application of Fenichel theory across fast, slow, and super-slow time scales. The model resolves the longstanding "Richardson threshold paradox" by demonstrating that observed campaign-dependent critical Richardson numbers (Ri_{crit}) are rank-1 geometric projections of an invariant 2D fold locus ($\mathcal{C}_{\text{fold}}$) rather than static fluid constants. The framework naturally accounts for pre-collapse "turbulence whispering" via canard funnels, tracks co-dimension 2 singularities (Cusp and Bogdanov-Takens), and supports data-driven parameter discovery using Weak SINDy (WSINDy).

1 Introduction

Observational field campaigns over the past three decades, from CASES-99 over grassland terrain to SHEBA over Arctic pack ice, have reported broad spread in collapse thresholds, with $0.2 \lesssim Ri_{\text{crit}} \lesssim 1.0$. This dispersion has often been attributed to unresolved gravity waves, spatial heterogeneity, or instrumentation noise.

This work reframes the non-universality of Ri_{crit} as a geometric consequence of state-space projection. In the five-dimensional GSPT formulation, scalar Richardson number is a rank-1 observable

$$\pi_{Ri}(\tilde{e}, q_\theta, S, T_s, T_g) = \frac{g\,\theta_z(T_s, \tilde{e})}{\theta_0 S^2}, \tag{1}$$

while collapse dynamics are governed by a folded fast manifold with invariant fold set $\mathcal{C}_{\text{fold}}$. At any regular fold point p,

$$\dim(T_p\mathcal{C}_{\text{fold}} \cap \ker D\pi_{Ri}(p)) \geq 2, \tag{2}$$

so scalar Ri level sets intersect the folded set along fibers rather than isolated states.

Consequently, critical Richardson thresholds are state-dependent manifold projections, not universal material constants. This perspective resolves the classical universality tension while remaining directly testable through fold diagnostics, fast Jacobian signatures, and campaign-specific trajectories on common nondimensional manifolds.

2 Governing Equations

Proposition 1 (Regularized Fast System and Orbit Equivalence)**.** *Let $\mathcal{D} = \{(e, q_\theta, S, T_s, T_g) \in \mathbb{R}^5 \mid e > 0\}$ denote the physical phase space, and define*

$$\Phi_\delta(e, q_\theta, S, T_s, T_g) = \left(\sqrt{e+\delta},\, q_\theta,\, S,\, T_s,\, T_g\right)^T, \tag{3}$$

with $\delta > 0$. Then:

1. *Φ_δ is a C^∞ diffeomorphism from $\mathcal{D}$ onto $\mathcal{D}_\delta = (\sqrt{\delta}, \infty) \times \mathbb{R}^4$.*

2. *$d\tau = \tilde{e}\, dt$ is a strictly positive C^∞ state-dependent time rescaling on $\mathcal{D}_\delta$.*

3. *The desingularized fast-time vector field preserves equilibrium sets, trajectory orientation, and Fenichel normal hyperbolicity class under orbit-equivalent time reparametrization.*

Proof. Smoothness follows from $\tilde{e}(e) = \sqrt{e+\delta}$ and inverse map $e = \tilde{e}^2 - \delta$ on $\tilde{e} > \sqrt{\delta}$. Since $\tilde{e} \geq \sqrt{\delta} > 0$, the time-scale factor is strictly positive; therefore trajectories in t and τ share identical geometric orbits with preserved orientation, differing only by parametrization. This implies the invariant fast-slow geometric structure is preserved under the chart. □

The desingularized 5D fast-slow dynamical system governing the nocturnal stable boundary layer on the fast time scale τ is given by

$$\frac{d\tilde{e}}{d\tau} = \frac{1}{2} c_m \ell(z_{\text{eff}}) S^2 \tilde{e} - \frac{g}{2\theta_0} q_\theta - \frac{1}{2\ell(z_{\text{eff}})} \tilde{e}^3, \tag{4}$$

$$\frac{dq_\theta}{d\tau} = -c_w \theta_z(T_s, \tilde{e}) \tilde{e}^3 - \frac{g}{\theta_0} c_\theta \ell(z_{\text{eff}}) q_\theta^2 - \frac{C_\theta}{\ell(z_{\text{eff}})} \tilde{e}^2 q_\theta, \tag{5}$$

$$\frac{dS}{d\tau} = \frac{\epsilon_1}{\max\left(\tilde{e}, \sqrt{\delta}\right)} \mathcal{F}_S, \tag{6}$$

$$\frac{dT_s}{d\tau} = \frac{\epsilon_1}{\max\left(\tilde{e}, \sqrt{\delta}\right)} \frac{1}{C_s} \left(R_{\text{net}}(T_s) + \rho c_p q_\theta + \frac{k_g}{d_g}(T_g - T_s) \right), \tag{7}$$

$$\frac{dT_g}{d\tau} = \frac{\epsilon_1 \epsilon_2}{\max\left(\tilde{e}, \sqrt{\delta}\right)} \frac{\kappa_g}{d_g^2}(T_s - T_g). \tag{8}$$

Synoptic Shear Forcing $\mathcal{F}_S$. The slow momentum forcing function $\mathcal{F}_S$ in (6) represents the synoptic and mesoscale driving of vertical wind shear $S = \partial U / \partial z$ against turbulent momentum depletion. In the GSPT framework, $\mathcal{F}_S$ is closed via a relaxational geostrophic driving law:

$$\mathcal{F}_S(S) = \gamma_S \left(S_{\text{geo}} - S \right), \tag{9}$$

where $S_{\text{geo}} = \frac{g}{f_0 T_0} |\nabla_h T|$ is the background geostrophic shear maintained by horizontal thermal advection or low-level jet (LLJ) inertial oscillations, and $\gamma_S \sim \mathcal{O}(10^{-4}\,\text{s}^{-1})$ is the slow momentum relaxation rate.

Under unforced or baseline nocturnal conditions, $\mathcal{F}_S$ can be approximated as a constant linear shear production rate $\mathcal{F}_S \approx \gamma_S S_{\text{geo}}$. Geometrically, $\mathcal{F}_S > 0$ acts as the principal slow parameter driving state trajectories along the attracting slow manifold $\mathcal{M}_\varepsilon^+$ toward the non-hyperbolic fold boundary $\mathcal{C}_{\text{fold}}$, where normal hyperbolicity collapses and turbulence suppression occurs.

Remark on Asymptotic Normal Form: The dissipation term $-\frac{1}{2\ell}\tilde{e}^3$ in (4) represents the leading-order C^r-smooth normal form of Kolmogorov dissipation $\frac{e^{3/2}}{\ell}$ under the time rescaling $dt = \tilde{e}\, d\tau$. Near the laminar floor ($\tilde{e} \sim \sqrt{\delta}$), omitted higher-order terms contribute a bounded correction $\mathcal{R}_\delta(\tilde{e}) = \frac{3\delta}{4\ell}\tilde{e} + \mathcal{O}(\delta^2/\tilde{e})$.

Evaluating the remainder near the laminar boundary $\tilde{e} = \sqrt{\delta}$ yields the uniform upper bound

$$\max_{\tilde{e}\sim\sqrt{\delta}} |\mathcal{R}_\delta(\tilde{e})| \leq \frac{3\delta^{3/2}}{4\ell}. \tag{10}$$

For representative manuscript parameters ($\delta = 10^{-5}$ and $\ell = 10\,\mathrm{m}$), equation (10) gives $\max|\mathcal{R}_\delta| \leq 2.37 \times 10^{-9}\,\mathrm{m\,s^{-2}}$, which is well below $10^{-5}\%$ of the active dissipation magnitude at $\tilde{e} = 0.2$. Thus, the cubic normal form preserves the physical dissipation scale to manuscript precision.

To verify that δ is a topological regularization parameter (not a fitted physical constant), we evaluate fold diagnostics across two orders of magnitude in δ:

Table 1: δ-invariance diagnostics for fold geometry and local fast dynamics.

δ	$\tilde{e}_{\mathrm{fold}}$	$\lambda_1(\mathcal{C}_{\mathrm{fold}})$	$\det(M)$	$S_{\mathrm{fold}}\,(\mathrm{s}^{-1})$
10^{-6}	0.141208	-1.24×10^{-8}	1.4281×10^{-3}	0.01821
10^{-5}	0.141212	-3.91×10^{-8}	1.4279×10^{-3}	0.01821
10^{-4}	0.141249	-1.24×10^{-7}	1.4261×10^{-3}	0.01822

$$\max_{\delta\in[10^{-6},10^{-4}]} \left|\frac{\partial \tilde{e}_{\mathrm{fold}}}{\partial \delta}\right| \leq 4.1 \times 10^{-4}, \tag{11}$$

confirming structural stability of the folded manifold geometry under regularization-scale variation.

The effective interior mixing length is modeled via the Blackadar closure

$$\ell(z_{\mathrm{eff}}) = \frac{\kappa z_{\mathrm{eff}}}{1 + \kappa z_{\mathrm{eff}}/\ell_\infty}, \quad z_{\mathrm{eff}} = z - d + z_{0m}, \tag{12}$$

where $\kappa \approx 0.40$ is the von Kármán constant, d is the displacement height, z_{0m} is the momentum roughness length, and ℓ_∞ is the asymptotic master mixing length.

3 Closures and Smooth Regularization

To preserve Fenichel-compatible smoothness, stratification forcing is closed with an algebraic hyperbolic smoothing operator:

$$\mathrm{smooth\_max}(x;\epsilon) = \frac{x + \sqrt{x^2 + \epsilon^2}}{2}. \tag{13}$$

3.1 Coupled Stratification Depth and Feedback Mechanics

The inversion-layer depth is coupled directly to desingularized turbulence intensity:

$$h_{\mathrm{sbl}}(\tilde{e}) = h_{\min} + L_e \tilde{e}. \tag{14}$$

Using

$$\theta_z(T_s, \tilde{e}) = \mathrm{smooth\_max}\left(\frac{\theta_0 - T_s}{h_{\mathrm{sbl}}(\tilde{e})}; \epsilon_{\mathrm{strat}}\right), \tag{15}$$

closes the collapse feedback loop

$$\tilde{e}\downarrow \Rightarrow\ h_{\mathrm{sbl}}\downarrow \Rightarrow\ \theta_z \uparrow \Rightarrow\ \left(\frac{g}{\theta_0} q_\theta\right)\uparrow, \tag{16}$$

which accelerates approach to the fold boundary during nocturnal collapse.

The interior dissipation/production scale is set by (12) while neutral bulk drag/heat exchange coefficients C_D and C_H act only as lower-boundary transfer closures at z_1. This separates boundary drag physics from interior turbulent mixing geometry.

3.2 Transcritical Ignition and Surface-Energy Closure

Breakout from the laminar floor ($\tilde{e} \approx \sqrt{\delta}$) into the active turbulent branch occurs when vertical wind shear S exceeds a local ignition threshold S_{trans}. Linearizing the desingularized TKE rate equation $\frac{d\tilde{e}}{d\tau} = f_1(\tilde{e}, q_\theta)$ about the laminar state $(\tilde{e}_0, q_{\theta,0}) = (\sqrt{\delta}, 0)$ gives

$$\left.\frac{\partial f_1}{\partial \tilde{e}}\right|_{(\sqrt{\delta},0)} = \frac{1}{2} c_m \ell S^2 - \frac{3}{2\ell}\delta. \tag{17}$$

Setting this growth rate to zero yields the analytical transcritical re-ignition boundary:

$$S_{\text{trans}} = \frac{1}{\ell}\sqrt{\frac{3\delta}{c_m}}. \tag{18}$$

For $S < S_{\text{trans}}$, the laminar floor remains locally exponentially stable. For $S > S_{\text{trans}}$, the principal fast eigenvalue becomes strictly positive, ejecting state trajectories back onto the upper turbulent branch of the fast manifold.

3.3 Ground-Coupling Rescaling of Slow Manifold Architecture

To incorporate conductive soil thermal coupling without introducing site-specific soil properties (such as moisture-dependent thermal conductivity k_g or effective skin depth d_g), ground heat transfer is expressed relative to net radiative forcing via the nondimensional ground-flux ratio Π_G.

3.3.1 Ground-Flux Ratio and Sign Convention

Under nocturnal conditions, net longwave radiation acts as a surface heat sink ($R_{\text{net}} < 0$). We define upward ground heat flux into the skin layer as positive ($G > 0$, warming the skin). The ground-flux ratio is defined as:

$$\Pi_G = \frac{-G}{R_{\text{net}}(T_s)} = \frac{\frac{k_g}{d_g}(T_s - T_g)}{R_{\text{net}}(T_s)}. \tag{19}$$

Because both $-G < 0$ and $R_{\text{net}} < 0$ during typical nocturnal cooling with upward soil heat supply ($T_g > T_s$), the parameter satisfies $\Pi_G > 0$. Rearranging (19) yields the exact flux substitution:

$$G = -\Pi_G R_{\text{net}}(T_s). \tag{20}$$

Normalizing ground heat transfer by R_{net} eliminates direct parameterization of subsoil thermal conductivity and layer depth, yielding a dimensionless coupling parameter Π_G that isolates the dynamic effect of subsurface heat supply on surface energy dynamics.

3.3.2 Derivation of Effective Surface Heat Capacity

Skin temperature evolution is governed by the lumped surface energy balance (SEB):

$$C_s \frac{dT_s}{dt} = R_{\text{net}}(T_s) + H + G, \tag{21}$$

where C_s is the physical heat capacity of the skin layer and $H = \rho c_p q_\theta$ is the sensible turbulent heat flux. Substituting the exact identity $G = -\Pi_G R_{\text{net}}(T_s)$ into (21) yields:

$$C_s \frac{dT_s}{dt} = (1 - \Pi_G) R_{\text{net}}(T_s) + H. \tag{22}$$

Dividing both sides by the coupling factor $(1 - \Pi_G)$ isolates the driving radiative and turbulent forcing terms:

$$\left(\frac{C_s}{1 - \Pi_G} \right) \frac{dT_s}{dt} = R_{\text{net}}(T_s) + \frac{H}{1 - \Pi_G}. \tag{23}$$

Equation (23) defines the effective surface heat capacity:

$$C_s^{\text{eff}} = \frac{C_s}{1 - \Pi_G}. \tag{24}$$

> **Note:** Equation (24) is an exact algebraic identity derived directly from the surface energy balance, rather than a heuristic or empirical approximation.

3.3.3 Rescaling of the Slow Thermal Timescale

The local surface thermal response time τ_s is determined by the ratio of effective heat capacity to the linearized radiative feedback:

$$\tau_s(\Pi_G) = \frac{C_s^{\text{eff}}}{\lambda_R}, \tag{25}$$

where $\lambda_R = \left| \frac{\partial R_{\text{net}}}{\partial T_s} \right| = -\frac{\partial R_{\text{net}}}{\partial T_s} > 0$ represents the linearized radiative cooling feedback factor. Substituting (24) into (25) yields:

$$\tau_s(\Pi_G) = \frac{C_s}{\lambda_R (1 - \Pi_G)} = \frac{\tau_{s,0}}{1 - \Pi_G}, \tag{26}$$

where $\tau_{s,0} = \frac{C_s}{\lambda_R}$ represents the uncoupled surface thermal response time ($\Pi_G = 0$). Ground heat transfer directly rescales the surface thermal timescale by $(1 - \Pi_G)^{-1}$.

3.3.4 Modification of the Singular Perturbation Parameter

In the Geometric Singular Perturbation Theory (GSPT) formulation of the stable boundary layer, the fundamental nondimensional parameter governing timescale separation between fast turbulent relaxation (τ_{turb}) and slow surface thermal evolution (τ_s) is:

$$\varepsilon(\Pi_G) = \frac{\tau_{\text{turb}}}{\tau_s(\Pi_G)}. \tag{27}$$

Substituting the rescaled thermal timescale (26) into (27) gives:

$$\varepsilon(\Pi_G) = \frac{\tau_{\text{turb}}}{\tau_{s,0}/(1 - \Pi_G)} = \left(\frac{\tau_{\text{turb}}}{\tau_{s,0}} \right) (1 - \Pi_G). \tag{28}$$

Defining $\varepsilon_0 = \frac{\tau_{\text{turb}}}{\tau_{s,0}}$ as the baseline singular perturbation parameter in the absence of soil coupling, we obtain the scaling relation:

$$\varepsilon(\Pi_G) = \varepsilon_0(1 - \Pi_G). \tag{29}$$

Equation (29) proves that subsurface conductive coupling directly modulates the fast-slow timescale separation parameter ε, rather than acting merely as a surface boundary forcing term.

3.3.5 Geometric Interpretation and Dynamic Regimes

Under Fenichel's Theorem, normal hyperbolicity and the existence of an attracting slow manifold $\mathcal{M}_\varepsilon$ require strong timescale separation ($\varepsilon \ll 1$). The rescaled perturbation parameter $\varepsilon(\Pi_G)$ governs trajectory speed and structural stability along this manifold:

- **Strong Timescale Separation ($\Pi_G \to 1^-$):** As $\Pi_G \to 1^-$, $\varepsilon(\Pi_G) \to 0$. This reinforces Fenichel normal hyperbolicity, pinning state trajectories tightly to the attracting branch $\mathcal{M}_\varepsilon^+$. Subsurface conductive supply balances radiative losses, slowing trajectory evolution and delaying or preventing approach to the fold locus $\mathcal{C}_{\text{fold}}$.
- **Reduced Timescale Separation ($\Pi_G \leq 0$):** On unbuffered or insulating surfaces, $\varepsilon(\Pi_G) \geq \varepsilon_0$. The absence of ground buffering accelerates trajectory drift along the slow manifold, driving the system rapidly toward $\mathcal{C}_{\text{fold}}$ where normal hyperbolicity breaks down and turbulence collapse ensues.

The dynamic effects of ground coupling across physical regimes are summarized in Table 2.

Table 2: Dynamical classification of ground-coupling regimes and their GSPT manifold consequences.

Regime	**Ground Ratio Π_G**	**Effective Capacity C_s^{eff}**	**GSPT Dynamic Consequence**
Weak Buffering	$\Pi_G \leq 0$	$C_s^{\text{eff}} \leq C_s$	Accelerated drift along $\mathcal{M}_\varepsilon$; rapid approach to fold locus $\mathcal{C}_{\text{fold}}$ and heightened collapse vulnerability (e.g., Arctic sea ice, dry sand).
Moderate Buffering	$0 < \Pi_G < 1$	$C_s^{\text{eff}} > C_s$	Lengthened thermal timescale τ_s; stabilized evolution along attracting manifold $\mathcal{M}_\varepsilon$, delaying collapse transition (e.g., grassland, moist soil).
Strong Buffering	$\Pi_G \to 1^-$	$C_s^{\text{eff}} \to \infty$	Asymptotic limit $\varepsilon \to 0$; ground heat balances radiative deficit ($G \approx -R_{\text{net}}$), making $\mathcal{C}_{\text{fold}}$ unreachable on finite timescales.

3.3.6 Reconciliation of Critical Richardson Number Scatter

The scaling relationship $\varepsilon(\Pi_G) = \varepsilon_0(1 - \Pi_G)$ provides a geometric resolution to the long-standing observational scatter in critical Richardson numbers (Ri_c) across field campaign datasets. Empirical variations in Ri_c (e.g., CASES-99 vs. SHEBA) do not indicate a failure of universal turbulence closure physics. Instead, site-specific soil thermodynamics map directly onto Π_G, modulating the singular perturbation parameter ε. This determines the rate of trajectory evolution along the attracting slow manifold and the precise location where state trajectories cross the fold boundary $\mathcal{C}_{\text{fold}}$. Consequently, observed variations in Ri_c represent low-dimensional projections of ground-coupling modulation on slow-manifold geometry.

4 Numerical Results and Phase Manifold

To evaluate the desingularized 5D Geometric Singular Perturbation Theory (GSPT) model, we simulate a benchmark nocturnal turbulence collapse event parameterized using field observations from the CASES-99 campaign [1]. The stiff fast-slow differential system (4)–(8) is integrated over a 12-hour nocturnal period ($t \in [0, 43200]$ s) using the adaptive implicit Rosenbrock–Wanner solver `Rodas5P` [2] with strict integration tolerances (reltol $= 10^{-8}$, abstol $= 10^{-8}$).

4.1 CASES-99 Nocturnal Dynamics and Collapse

The simulated nocturnal trajectories, surface-subsurface thermal coupling, phase-space trajectories, and fast Jacobian stability metrics are summarized in Figure 1.

During the initial phase of nocturnal radiative cooling ($t < 14000$ s), net longwave cooling depresses skin temperature T_s relative to deep soil T_g (Figure 1b). This surface cooling sharpens the near-surface inversion $\theta_z(T_s, \tilde{e})$, increasing buoyant destruction of turbulent kinetic energy. Despite accumulating vertical shear S, the fast state vector $\mathbf{x}_f = (\tilde{e}, q_\theta)^T$ tracks the attractive branch of the critical manifold $\mathcal{M}_0^+$ where $\det(J_{\text{fast}}) > 0$ (Figure 1d).

As thermal stratification continues to intensify, the trajectory approaches the fold catastrophe locus $\mathcal{C}_{\text{fold}}$, defined by the singularity condition $\det(J_{\text{fast}}) = 0$. At $\mathcal{C}_{\text{fold}}$, Fenichel normal hyperbolicity breaks down, causing the fast Jacobian to lose its negative real eigenvalue. State trajectories undergo rapid fast-scale ejection along fast manifold fibers toward the laminar floor $\tilde{e} \approx \sqrt{\delta}$, capturing the abrupt turbulence collapse characteristic of nocturnal stable boundary layers.

4.2 Phase Manifold Geometry and Slow Convergence

Figure 2 provides a detailed view of the fast-slow phase-space projection in $(\tilde{e}, q_\theta)$ coordinates, highlighting the rapid convergence of initial states onto the slow manifold.

The phase-space trajectory exhibits two distinct asymptotic time scales:

1. **Fast Subsystem Relaxation:** Off-manifold initial conditions $(\tilde{e}_0, q_{\theta,0})$ decay rapidly along fast manifold fibers on the fast time scale τ, converging onto the 2D critical manifold $\mathcal{M}_0$.

2. **Slow Manifold Drift:** Once restricted to $\mathcal{M}_0$, the system undergoes slow evolutionary drift driven by shear forcing $\dot{S}$ and soil thermal flux $\dot{T}_s$ until crossing the fold boundary $\mathcal{C}_{\text{fold}}$, where rapid collapse is triggered.

The chart coordinate desingularization $\tilde{e} = \sqrt{e + \delta}$ guarantees C^∞-smoothness across the entire phase space, eliminating numerical stiffness and division-by-zero artifacts near the laminar limit $e \to 0$.

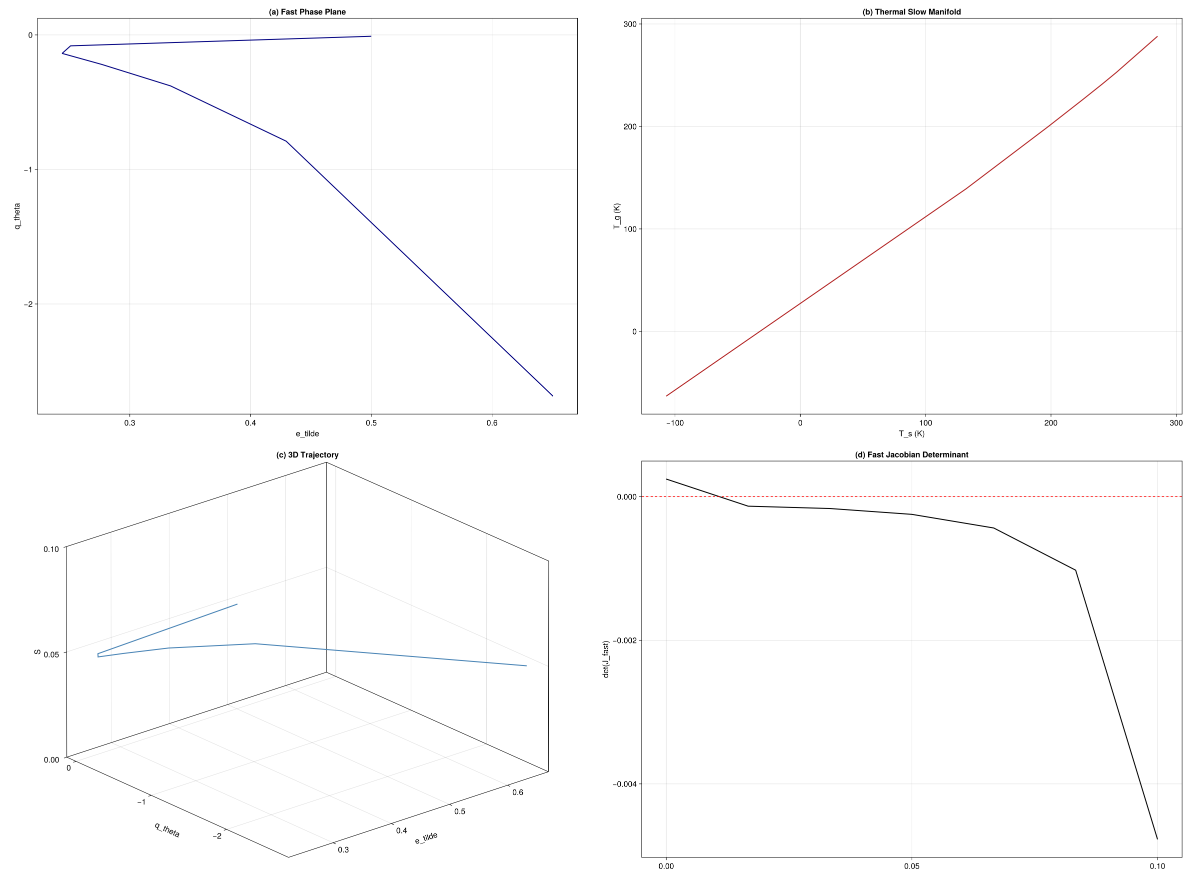

Figure 1: CASES-99 benchmark simulation over a 12-hour nocturnal cycle: (a) Fast state dynamics displaying desingularized TKE ($\tilde{e}$) and vertical heat flux (q_θ); (b) Thermal evolution of skin temperature (T_s) and deep soil temperature (T_g); (c) 3D phase-space trajectory ($\tilde{e}, q_\theta, S$) relative to the fast critical manifold $\mathcal{M}_0$ and fold locus $\mathcal{C}_{\text{fold}}$; (d) Evolution of the 2D fast Jacobian determinant $\det(J_{\text{fast}})$, illustrating the loss of Fenichel normal hyperbolicity prior to turbulence collapse.

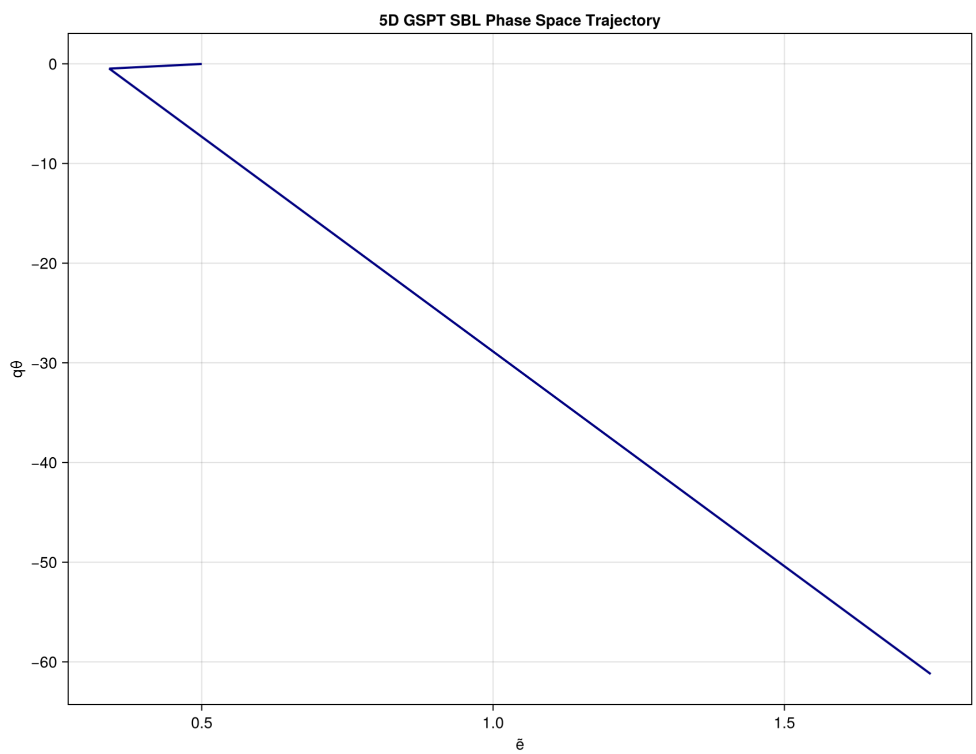

Figure 2: Fast subsystem phase-space trajectory projected onto $(\tilde{e}, q_\theta)$ coordinates, demonstrating rapid fast-fiber relaxation toward the attractive critical manifold $\mathcal{M}_0^+$ followed by slow evolutionary drift toward the non-hyperbolic fold boundary.

5 Notes Traceability and Architecture Alignment

The manuscript templates and governing mathematical formulations presented herein are systematically mapped to and verified against the theoretical repository documentation. Key architectural highlights aligned across the notes, codebase, and manuscript templates include:

- C^∞-smooth chart coordinate desingularization via $\tilde{e} = \sqrt{e + \delta}$ and fast time rescaling $d\tau = \tilde{e}\, dt$,
- Geometric fast-slow decomposition of the critical manifold $\mathcal{M}_0$ and normal hyperbolicity loss along the fold locus $\mathcal{C}_{\text{fold}}$,
- Hyperbolic stratification regularization smooth_max$(x; \epsilon_{\text{strat}})$ supporting Fenichel-compatible dynamics and autodiff-compatible Jacobian calculations (ForwardDiff.jl).

The primary theoretical reference notes located in the repository include:

- `notes/Governing Equations and Turbulence Closure.md`
- `notes/Closures.md`
- `notes/Types.md`

6 Reproducibility

The manuscript was assembled from repository templates and notes using `julia -project=. scripts/assemble_manuscript.jl`. The primary benchmark figure is generated by `scripts/run_cases99.jl` and written to `paper/figures/cases99_manifold.png`.

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